

On the biquadratic character of the number “two”

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From a letter of Dirichlet to Mr. Stern in Göttingen, Crelle's Journal for Pure and Applied Mathematics, vol. 57, pp. 187-188.

On the biquadratic character of the number “two”.

Let p be a prime number of the form $4n + 1$ and

$$p = a^2 + b^2,$$

where a is odd. From this equation and from the equation following from it,

$$2p = (a + b)^2 + (a - b)^2,$$

one obtains at once, using Legendre's generalized symbol,

$$\left(\frac{p}{a}\right) = 1, \quad \left(\frac{2p}{a+b}\right) = 1$$

or

$$\left(\frac{p}{a+b}\right) = \left(\frac{2}{a+b}\right),$$

and then, by known theorems,

$$\left(\frac{a}{p}\right) = 1, \quad \left(\frac{a+b}{p}\right) = (-1)^{\frac{1}{8}[(a+b)^2-1]},$$

or, what is the same thing,

$$a^{\frac{1}{2}(p-1)} \equiv 1, \quad (a+b)^{\frac{1}{2}(p-1)} \equiv (-1)^{\frac{1}{8}[(a+b)^2-1]} \pmod{p}.$$

On the other hand, from

$$(a+b)^2 \equiv a^2 + b^2 + 2ab \equiv 2ab \pmod{p}$$

one obtains, by raising to the power $\frac{1}{4}(p-1)$,

$$(a+b)^{\frac{1}{2}(p-1)} \equiv 2^{\frac{1}{4}(p-1)} a^{\frac{1}{4}(p-1)} b^{\frac{1}{4}(p-1)} \pmod{p},$$

or, if one sets

$$b \equiv af$$

and takes account of the two last congruences,

$$(-1)^{\frac{1}{8}[(a+b)^2-1]} = (-1)^{\frac{1}{8}(p-1+2ab)} \equiv 2^{\frac{1}{4}(p-1)} f^{\frac{1}{4}(p-1)} \pmod{p}.$$

Now, since

$$a^2 + b^2 = p, \quad b^2 \equiv a^2 f^2,$$

therefore

$$f^2 \equiv -1 \pmod{p},$$

and hence

$$(f^2)^{\frac{1}{8}(p-1+2ab)} = f^{\frac{1}{4}(p-1)} \cdot f^{\frac{1}{2}ab} \equiv 2^{\frac{1}{4}(p-1)} \cdot f^{\frac{1}{4}(p-1)} \pmod{p}.$$

Dividing by $f^{\frac{1}{4}(p-1)}$ gives

$$2^{\frac{1}{4}(p-1)} \equiv f^{\frac{1}{2}ab} \pmod{p},$$

which is transformed into the result given by Gauss in § 24 of his *theoria residuorum biquadraticorum* if one multiplies by $a^{\frac{1}{2}ab}$ and again writes b for af .

Göttingen, 21 January 1857.